Econ 2 - Lecture 5 - 1/21/26

Lecture Quiz 3 → Quick turnaround → Released today, due Monday

Discussion Activity # 2 this week!

Today: GDP Calculation + CPI Introduction

Next Week: CPI + Unemployment

Monday, February 2nd = Midterm Exam (30 MC Questions)

Lots of Resources! "Additional Practice Problems" folder
Review Session Questions (foundation)
Old Midterm

Best Resource?
Write own Questions!

Last Class:

Product	2009 Quantity	2009 Price	2016 Quantity	2016 Price
Food	1000	1	1500	2
Housing	100	100	150	150
Movies	500	5	1000	10

$$\text{Nominal GDP}_{2009} = Q_{09} \times P_{09} \rightarrow \text{sum for all goods}$$

$$= Q^F_{09} \times P^F_{09} + Q^H_{09} \times P^H_{09} + Q^M_{09} \times P^M_{09}$$

$$= 1000 \times 1 + 100 \times 100 + 500 \times 5 = \$13{,}500$$

$$\text{Nominal GDP}_{2016} = Q_{2016} \times P_{2016}, \text{ sum for all goods}$$

$$= 1500 \times 2 + 150 \times 150 + 1000 \times 10$$

$$= 35{,}500$$

Nominal GDP increased a lot $\Rightarrow$ Prices $\uparrow$, Quantity $\uparrow$

13,500 $\swarrow$ $\searrow$ 35,500

Price increase $\Rightarrow$ is not production increase

Real GDP: Separate price increase from quantity increase

Calculate GDP using the same set of prices every year

Step 1: Define a base year = year we fix prices

→ Assume that base year is 2009 (can be any year)

→ Base year cannot change

Step 2: Calculate GDP, but only use base year prices

$$\text{Real GDP}_{2016} = Q^{F}_{16} \times P^{F}_{BY} + Q^{H}_{16} \times P^{H}_{BY} + Q^{M}_{16} \times P^{M}_{BY}$$

BY = Base Year = 2009

$$= 1500 \times 1 + 150 \times 100 + 1000 \times 5$$

$$\text{Real GDP}_{2016} = \$21,500$$

If prices did not change between 2009 + 2016, the nominal GDP would have been 21,500 NOT 35,500

21,500 ↗ Real GDP

35,500 ↗ Nominal GDP

Note: $\text{Real GDP}_{CY} = Q_{CY} \times P_{BY}$ (multiply + sum)

CY = current year

$$\text{Real GDP}_{CY=BY} = \text{Nominal GDP}_{CY=BY}$$

Separate price increases from quantity increases

→ GDP Deflator ⇒ Weighted Price Ratio

$$\text{GDP Deflator}_{CY} = \frac{\text{Nominal GDP}_{CY}}{\text{Real GDP}_{CY}} \times 100$$

$$= \frac{Q_{CY} \times P_{CY}}{Q_{CY} \times P_{BY}} \times 100$$

$$\text{GDP Deflator}_{BY=2009} = \frac{Q_{2009} \times P_{2009}}{Q_{2009} \times P_{BY=2009}} \times 100 = 100$$

Base Year GDP Deflator = 100

$$\text{GDP Deflator}_{2016} = \frac{35,500}{21,500} \times 100 = 165$$

Price of the "average good" increased from 100 in base year to 165 in 2016

$\text{Inflation Rate}_{09,16} = 65\%$

Convert GDP Deflator → inflation rate

$$\text{Percentage Change in GDP Deflator}_{Yr1,Yr2} = \frac{\text{GDP Def}_{Yr2} - \text{GDP Def}_{Yr1}}{\text{GDP Def}_{Yr1}} \times 100$$

$$= \frac{165-100}{100} \times 100 = \frac{65}{100} \times 100 = 65\%$$

$\text{GDP Deflator}_{17} = 172$

$$\text{Inflation Rate}_{16,17} = \frac{\text{GDP Def}_{17} - \text{GDP Def}_{16}}{\text{GDP Def}_{16}} \times 100$$

$$= \frac{172-165}{165} \times 100 = \frac{7}{165} \times 100 = 4.24\%$$

Alternative Measures of GDP

Value-Added GDP

Best Buy computer → $400 final sale

→ $50 in raw materials ⇒ $50

→ Raw materials
→ Parts → $150
→ Add $100 in value

→ Parts → $350 Dell Computer
→ Add $200 in Value

→ Central Location → $400 Purchase
→ Add $50 in Value

→ do not add up intermediate goods
→ Difference in value of production process
⇒ final contribution to GDP

⇒ 50 + 100 + 200 + 50 = $400

Factor Payments GDP

Where does "profit" go? Income

$50 Made by Best Buy:
→ Labor income
→ Stockholder income
→ Rental Income
→ Interest on debt

Overall, GDP = Y = C + I + G + NX
= Firm Profit (Value-Added)
= Income (Factor Payments)

Goal #1: High Standard of Living

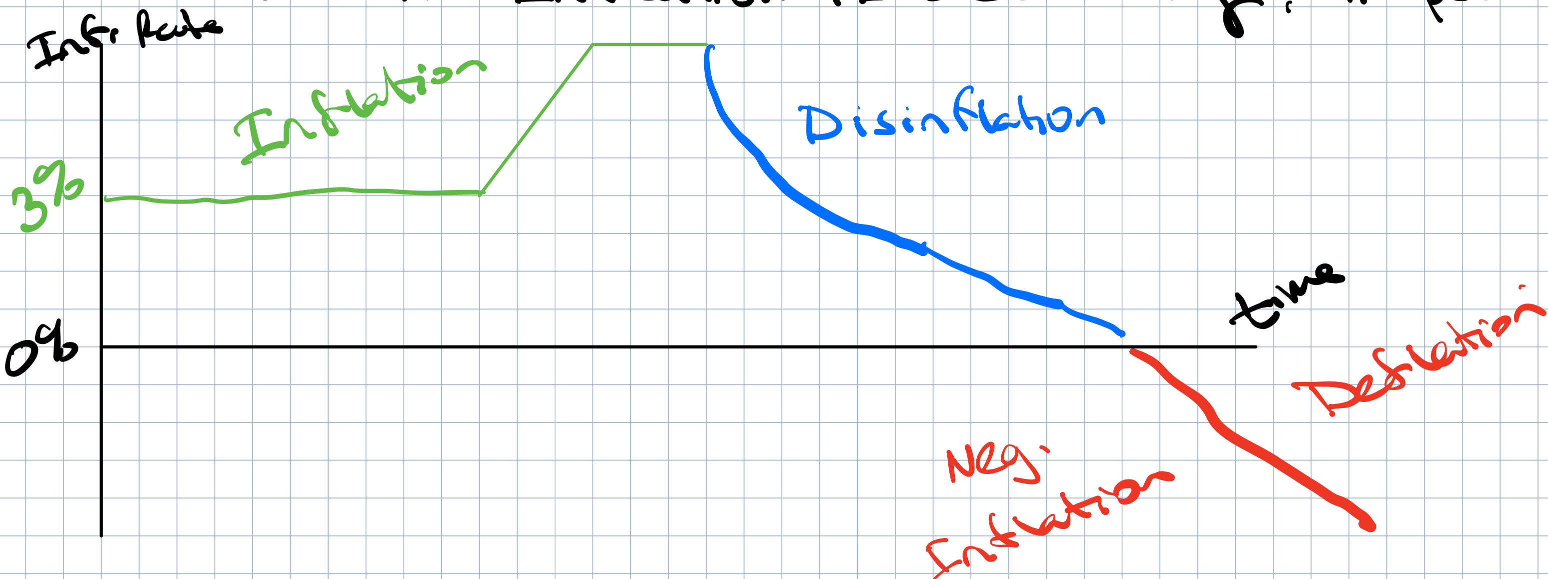
Goal #2: Stable Prices

GDP Deflator => measure of prices using the GDP

Inflation = Prices are rising

Deflation = Negative Inflation = Prices are falling

Disinflation = Inflation is decreasing, still positive



Stable Price Goal → Federal Reserve

Inflation Rate Target = 2%

Low & Predictable → Inflation Expectation

→ Anchored at 2%

Avoid deflation! Deflationary Spiral

If prices are expected to fall → Delay buying

→ Less Production → fewer jobs →

GDP Deflator measures inflation

Use all products in GDP

Do consumers care about prices of all goods in GDP?

Isolate price changes relevant to households

Consumer Price Index (CPI)

Only consider consumer goods in CPI